

Def. A continuous real-valued $f: D \rightarrow \mathbb{R}$ on a domain $D \subset \mathbb{C}$ is called harmonic if:
 f is partially twice continuously differentiable ($f \in C^2$) satisfying

$$f_{xx} + f_{yy} = 0. \quad (\text{It is called the Laplace's equation.})$$

Thm. Let $f = u + iv$ be an holomorphic on a domain D .

Then, both u and v are harmonic on D .

Proof. By the Cauchy-Riemann equation, $u_x = v_y$ and $u_y = -v_x$, so that

$$f' = u_x + i v_x = u_x - i u_y = v_y + i v_x$$

$$\begin{aligned} \hookrightarrow f'' &= u_{xx} - i u_{xy} = v_{yx} + i v_{xx} \\ &= u_{xx} - i u_{xy} = v_{xy} + i v_{xx} \\ &= u_{xx} - i v_{yy} = -u_{yy} + i v_{xx} \end{aligned} \quad \text{By Fubini's Theorem}$$

Therefore, $u_{xx} = -u_{yy}$ and $-v_{yy} = v_{xx}$.

Thm. Suppose that f is analytic on D satisfying $|f(z)|$ is constant on D .

Then f is constant in D .

Proof. Denote $f = u + iv$. By the assumption,

$$|u + iv| = C \text{ for all } z \in D, \text{ so } u^2 + v^2 = C^2.$$

Differentiating with respect to real and imaginary, then

$$2u u_x + 2v v_x = 2u u_y + 2v v_y = 0,$$

substituting the terms in Cauchy-Riemann equation, $u_x = v_y$ and $v_x = -u_y$.

$$u u_x - v u_y = u u_y + v u_x = 0, \text{ multiplying } u \text{ to the left side, then}$$

$$u u u_x - u v u_y = u u u_x + v v u_x = u_x (u^2 + v^2) = 0, \text{ so } u_x = 0.$$

(If $u_x \neq 0$, then $u^2 + v^2 = C^2 = 0$, it means f is identically zero).

Similarly, $u_y = v_x = v_y = 0$, so f' is zero, it gives the f is constant. $\square$

Another proof of: $|f|$ is constant $\Rightarrow f$ is constant.

Lemma. If f is real-valued and holomorphic on a domain D , then f is constant.

Proof. Given $z_0 \in D$,

$$\lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} \text{ exists by the hypothesis, so}$$

letting $h \rightarrow 0$ on pure real axis, then the derivative is also pure real, since the f can have real value. Similarly, letting $h \rightarrow 0$ on pure imaginary, the derivative must be pure imaginary. So, $f'(z_0)$ must be zero. Since z_0 was arbitrary, $f' \equiv 0$ on D , so f is constant. $\square$

Now, suppose that $|f| \equiv C$ for some $C \in \mathbb{C}$.

If $C=0$, then $f=0$, so there is nothing to prove.

If $C>0$, then $|f|^2 = f \bar{f} = C^2$, so $\bar{f} = \frac{C^2}{f}$ is analytic on D .

$$\text{Therefore, } \operatorname{Re} f = \frac{f + \bar{f}}{2} \text{ and } \operatorname{Im} f = \frac{f - \bar{f}}{2i}$$

both are holomorphic, so constant, by the Lemma above. $\square$

And, using the Lemma, show that a fun result:

Thm. If image of f lies on a line in $\mathbb{C}$ on a domain D , then f is constant.

Proof. Let

Thm.

If $U: D \rightarrow \mathbb{R}$ is harmonic on a simply connected domain D , then there exists an analytic function f on D satisfying $\operatorname{Re}(f) = U$ on D .

Proof. Define $g(z) = U_x(z) - i \cdot U_y(z)$. Then, since the U is harmonic and so $U \in C^2$,

$$U_{xx} = -U_{yy}, \quad \text{and } U_{xy} = U_{yx} \text{ by the Fubini's Theorem, } \dots (*)$$

Actually, the $(*)$ is a Cauchy-Riemann equation for g , so g is analytic on D , since $U \in C^2$. Let $z_0 \in D$ be fixed. Define

$$F(z) = \int_{z_0}^z g(u) du \quad \text{on } D.$$

So F is defined as integral of an analytic function, so F is analytic. And

$$\begin{aligned} F'(z) &= g(z) = U_x(z) - i \cdot U_y(z) \\ &= \frac{\partial}{\partial x} \operatorname{Re} F(z) - i \cdot \frac{\partial}{\partial y} \operatorname{Re} F(z) \end{aligned}$$

So that $\operatorname{Re} F(z) = U(z) + C$ for some constant $C \in \mathbb{R}$, therefore

$$f(z) = F(z) - C = \int_{z_0}^z (U_x(u) - i \cdot U_y(u)) du - C$$

is we desired. $\square$

Example. Let $U(x, y) = \sin x \cdot \cosh y + \cos x \cdot \sinh y + x^2 - y^2 + 2xy$.

By the theorem, there is analytic $f = U + iV$, so

$$f'(z) = U_x - iU_y = \cos x \cdot \cosh y - \sin x \sinh y + 2x + 2y$$

$$\text{By definition, } \cos(iy) = \frac{e^{i \cdot iy} + e^{-iy}}{2} = \frac{-i(\sin x \cdot \sinh y + \cos x \cdot \cosh y - 2y + 2x)}{2} = \cosh(y) \text{ and } \sin(iy) \cdot (-i) = \sinh y,$$

So we can simplify

$$f'(z) = \cos x \cdot \cos(iy) + i \cdot \sin x \cdot \sin(iy) + 2x - 2y$$

$$= i \cdot (-i \cdot \sin x \cdot \sinh y) + \cos x \cdot \cos(iy) - 2y + 2x$$

$$= (1 - i)(\cos x \cdot \cos(iy) - \sin x \cdot \sin(iy) + 2z)$$

$$= (1 - i) \cdot (\cos z + 2z)$$

$$\text{So, } f(z) = (1 - i) \cdot (\sin z + z^2) + C.$$

Thm.

A harmonic and bounded function must be a constant.

Proof. Suppose that $u(z)$ is harmonic and bounded.

By the theorem, there is an analytic $f: \mathbb{C} \rightarrow \mathbb{C}$ such that $\operatorname{Re} f = u$.

Then, $g(z) = e^{f(z)}$ satisfies $|g(z)| = e^{u(z)}$

is bounded, since u is bounded. Therefore, by the Liouville's theorem, the g is constant, also f , so also u . $\square$